
mini-spectrometer

Type number : C12880MA

INSTRUCTION MANUAL



Read this instruction manual carefully before using this product.

Improper handling and operation that does not comply with the instructions in this manual may result in serious accident including injury or death.

Keep this instruction manual handy so that you can easily refer to it when needed.

使用本产品前请仔细阅读本说明书。

不按照本手册的说明进行处理和操作可能会导致严重的事故，包括人身伤害甚至死亡。

请将本说明书放在手边，以便在需要时进行参考。



Introduction / 概述

Please read carefully / 请仔细阅读

- Please carefully read the "Precautions" before using.
使用前请仔细阅读“注意事项”。
- Please read this manual carefully and use the product correctly.
请仔细阅读本说明书，并正确使用本产品。
- Please keep this manual and read when it is necessary.
请妥善保管本说明书，在必要时进行阅读。

This manual uses the following safety alert symbols and signal words.
本说明书使用以下安全警示符号和信号词。

 DANGER/危险	Indicates an imminently hazardous situation which if not avoided, will result in death or serious injury. This signal word is to be limited to the most extreme situations. 用于警示紧急的危险情形，若不可避免，将会导致人员死亡或严重的人身伤害。这一信号词仅限于在最极端的情况下使用。
 WARNING/警告	Indicates a potentially hazardous situation which if not avoided, could result in death or serious injury. 用于警示潜在的危险情形，若不可避免，可能会导致人员死亡或严重的人身伤害。
 CAUTION/注意	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It is permitted for property damage only accidents. 用于警示潜在的危险情形，若不可避免，可能会导致人员受到轻微或中等伤害。这一信号词仅允许在发生财产损失事故的情况下使用。

	Indicates a note to help you get the best performance from the system. Read the contents of the note carefully to ensure correct and safe use. Failure to observe one of these notes might impair the performance of the system. 本注意事项用于帮助您从系统中获得最佳性能。请仔细阅读注意事项内容，以确保正确、安全地进行使用。未遵循其中任何一项注意事项都可能会影响系统性能。
	Indicates a cautionary item that should be obeyed when handling the system. Read the contents carefully to ensure correct and safe use. 表示操作本系统时应遵守的注意事项。请仔细阅读其中内容，以确保正确、安全地进行使用。
	Indicates an action that is forbidden. Read the contents carefully and be sure to obey them. 表示禁止的操作。请仔细阅读其中内容并予以遵守。
	Indicates a compulsory action or instruction. Read the contents carefully and be sure to obey them. 表示强制性操作或注意事项。请仔细阅读其中内容并予以遵守。

Safety consideration / 安全考虑

Make sure you follow the instructions / 确保你遵循指示

This product is intended for use in general electronic equipment (Measurement equipment, office equipment, information and communications equipment, home appliances, etc.). Do not use this product for special purposes (Nuclear control equipment, Aerospace equipment, Medical equipment and Transport equipment that directly affect human life, disaster prevention and safety equipment, etc.) that require extremely high reliability and safety, except as described in the individual product documentation. In order to use this product correctly and safely, make sure to follow the safety notices below when operating this product. Hamamatsu Photonics is neither responsible nor liable for damages caused by using this product contrary to the safety notices.

本产品适用于一般电子设备（测量设备、办公设备、信息通信设备、家用电器等）。除非个别产品文档中另有说明，否则请勿将本产品用于需要极高可靠性和安全性的特殊用途（核控制设备、航空航天设备、直接影响人类生命的医疗设备和运输设备、防灾和安全设备等）。为了正确、安全地使用本产品，操作产品时请务必遵守以下安全注意事项。对于违反安全注意事项使用本产品所造成的损害，Hamamatsu Photonics 不负任何责任。



WARNING / 警告

● In case an abnormality has occurred / 在发生异常情况时

Do not use this product if smoke, unusual odor, abnormal noise, or heat is generated during operation. Continuing operation may cause a fire or electrical shock. In this case, immediately turn off the PC (personal computer) connected to this product, and then unplug the USB cable. After making sure that the abnormal condition has disappeared, contact our sales office. Never attempt to repair this product yourself since it can be dangerous.

如果操作过程中出现烟雾、异味、异常噪音或发热情况，请勿使用本产品。继续操作可能会引发火灾或触电。在这种情况下，请立即关闭连接本产品的 PC（个人电脑），然后拔下 USB 线。确认异常情况消失后，请与我们的销售办公室联系。切勿尝试自行维修本产品，因为这样做可能十分危险。

● Do not use this product above its absolute maximum ratings

/使用本产品时，切勿超过其绝对最大额定值

Do not use this product above values stated in the absolute maximum ratings. Using this product beyond its absolute maximum ratings may cause damage or fires.

使用本产品时，切勿超过其标注的绝对最大额定值。使用本产品时超过其绝对最大额定值可能会造成产品损坏或导致火灾。

● Absolutely do not make mistaken connections / 严禁出现连接错误

Doing so may cause overcurrent, causing damage or fires in parts and devices.

这样做可能会引起过电流，从而导致零件和设备损坏或起火。

Precautions for use / 使用注意事项

Make sure you follow the instructions/ 确保你遵循指示

Before using this product, be sure to read the following precautions to ensure proper use.

使用本产品前，请务必阅读以下注意事项，确保正确使用。

- The product is a precision optical instrument. Do not subject the Product to excessive vibration or shock. Avoid operation and storage in dusty, watery, high temperature and humid environments.

该产品是一种精密光学仪器。不要使产品受到过度振动或冲击。避免在灰尘、潮湿、高温和潮湿的环境中操作和储存。

- When cleaning the case, rub it dry with a clean, soft cloth. Do not use organic solvents such as thinner or acetone.

清洁外壳时，用干净柔软的布擦干。不要使用有机溶剂，如稀释剂或丙酮。

- Use the camera within the range where the A/D count of each pixel is not saturated by the incident light. If saturation of A/D counts occurs, set a shorter accumulation time. If the accumulation time cannot be set short, the use of a light-reducing filter is recommended at the user's side.

在每个像素的 A/D 计数未被入射光饱和的范围内使用相机。如果 A/D 计数出现饱和，设定较短的累积时间。如果不能缩短累积时间，建议用户使用减光过滤器。

- Even in the absence of incident light, the product has an output consisting of an offset component and a dark output. Note that the offset component does not depend on the accumulation time, but the dark output does.

即使在没有入射光的情况下，产品也具有由偏置组件和暗输出组成的输出。注意，偏移组件不依赖于累积时间，但是暗输出依赖于累积时间。

- Window / 窗户

- Do not apply strong friction or pressure to prevent scratching or cracking of the window material. Avoid contact with sharp or hard objects and window materials.

不要施加强烈的摩擦或压力以防止刮伤或开裂的窗户材料。避免接触尖锐或坚硬的物体和窗户材料。

- Dust on the window material should be removed using an air blower. It is recommended that an ionizer be used in conjunction with an ionizer.

窗户上的灰尘应使用鼓风机清除。建议将电离器与电离器结合使用。

- If the window material is contaminated with oil, grease, or other contaminants that cannot be removed by air blow, gently wipe it off with a cotton swab or similar object moistened with ethyl alcohol to avoid scratching. Hard rubbing or repeated wiping may cause scratches, resulting in deterioration of electrical/optical characteristics and reliability.

如果窗户材料被油、油脂或其他空气吹不掉的污染物污染，用棉签或类似的物体用乙醇液轻轻擦拭，以免划伤。用力摩擦或反复擦拭可能会造成划痕，导致电气/光学特性和可靠性下降。

- Care should also be taken when packing and transporting equipment with the product incorporated to prevent dirt and scratches on the window material.

在包装和运输设备时也应注意与产品结合，以防止污垢和划伤窗户材料。

● Static electricity management / 静电管理

The products or packaging is marked with an electrostatic warning in accordance with the figure below to warn against static electricity.

产品或包装上按照下图标出静电警告以警告静电。



(1) Workplace and equipment, etc. / 工作场所和设备等

- The surface of the work desk should be grounded with a conductive mat (750 k Ω to 1 G Ω).
工作台表面应用导电垫 (750 k Ω ~1 G Ω) 接地。
- The floor of the work area should be grounded by using conductive flooring or by laying down conductive mats.
工作区域地面应使用导电地板或铺设导电垫接地。
- All manufacturing and inspection equipment should be grounded.
所有制造和检查设备均应接地。
- Humidity should be around 50 %. Low humidity tends to generate static electricity, while high humidity tends to absorb moisture.
湿度应在 50% 左右。低湿度容易产生静电，而高湿度容易吸收水分。

(2) Handling / 处理

- It is recommended that ionizers or similar devices be used to eliminate static when handling the product.

操作本产品时，建议使用静电消除器或类似装置消除静电。

- Wear antistatic clothing and conductive shoes (100 k Ω to 100 M Ω).

穿着防静电服和导电鞋(100 k Ω ~100 M Ω)。

- Wrist straps should be worn in direct contact with bare skin and grounded. Use a wrist strap with a protective resistance of 750 k Ω to 35 M Ω when measured with the strap in place. If the wrist strap does not have a protective resistance, there is a risk of electric shock due to electrical leakage, which is extremely dangerous. Also, use conductive finger sacks or gloves.

腕带应直接接触裸露皮肤并接地。请使用保护电阻为 750 k Ω ~ 35 M Ω （带上测量）的腕带。如果腕带没有保护电阻，则可能会因漏电而导致触电，非常危险。此外，请使用导电指套或手套。

- Tools such as tweezers for handling the product may also be electrically charged. Ground them as necessary. 用于操作产品的镊子等工具也可能带电。请根据需要将其接地。

- Use a soldering iron with an insulation resistance of 10 M Ω or higher and a grounded tip.

使用绝缘电阻为 10 M Ω 或更高且尖端接地的烙铁。

- If the product is induction-charged and comes into contact with a metal, excessive current may flow due to discharge, resulting in electrostatic breakdown. To prevent induction charging, keep objects (insulators such as plastic and vinyl, PC displays and keyboards, etc.) that may be electrically charged away from the product. Simply bringing them close to the product may cause the product to become induction-charged. If it is unavoidable to bring the product closer to such objects, use an ionizer or similar device to eliminate static from the objects that may be charged.

如果本产品带有感应电荷并接触金属，可能会因放电而流过过大的电流，导致静电击穿。为防止感应带电，请将可能带电的物体（例如塑料和乙烯基等绝缘体、PC 显示器和键盘等）远离本产品。仅仅将它们靠近本产品可能会导致本产品感应带电。如果无法避免，请使用静电消除器或类似装置消除可能带电物体上的静电。

- Friction may cause electrification of the product. If friction is unavoidable, use an ionizer to eliminate static electricity.

摩擦可能导致本产品带电。如果无法避免摩擦，请使用静电消除器消除静电。

- Peripheral devices must be grounded to prevent surges due to leakage voltage. Do not apply voltages exceeding the absolute maximum ratings from measuring instruments, etc. (especially when turning power on and off). If there is a possibility that a surge may be applied, use a filter (consisting of resistors and capacitors) for protection. Do not attach or remove any connectors, etc. connected to the power supply line or output line during operation.

外围设备必须接地，以防止漏电电压引起的浪涌。请勿施加超过测量仪器等绝对最大额定值的电压（尤其是在打开和关闭电源时）。如果可能施加浪涌，请使用滤波器（由电阻器和电容器组成）进行保护。请勿在运行过程中装卸连接到电源线或输出线的任何连接器等。

(3) Transportation, storage, packaging / 运输, 储存, 包装

- Insert the product into a conductive mat (shorting the leads) and place it in a conductive case. Do not use plastic, Styrofoam, etc., as vibration during transportation generates static electricity, which may cause destruction or deterioration of the product.

将产品插入导电垫（缩短引线）并将其放入导电盒中。不要使用塑料、泡沫塑料等，因为运输过程中的振动会产生静电，从而导致产品损坏或变质。

- Transport cases and storage shelves should be conductive.

运输箱和储存架应具有导电性。

- Do not store the product near equipment that generates high voltage or high electromagnetic fields.

请勿将产品存放在产生高压或高电磁场的设备附近。

- When packaging the product, short-circuit the leads to make them equipotential and use conductive materials.

Note: It is not necessary to take all of the above measures against static electricity. Take measures according to the degree of possible damage.

包装产品时，请将引线短路使其等电位，并使用导电材料。

注意：无需全部采取上述防静电措施。请根据可能造成的损害程度采取措施。

【Examples of anti-static measures / 防静电措施示例】

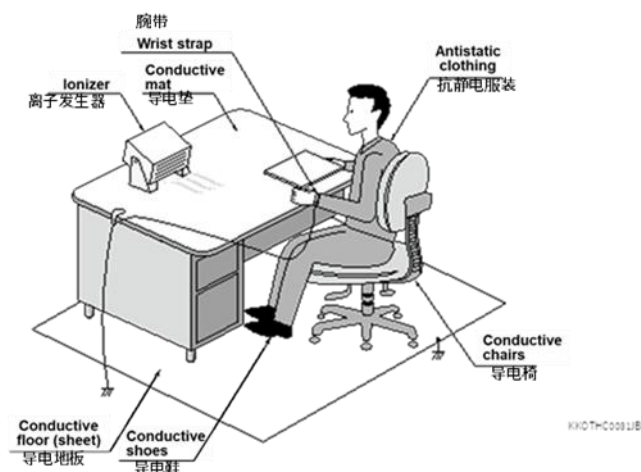


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1. Check package contents

After unpacking, make sure to check that there are the following components: In the unlikely event that the delivered products are incorrect, incomplete, or damaged, do not operate the products, and make sure to contact Hamamatsu Photonics or our distributor.

Packaged item list

	Contents	Photo	Description
1	C12880MA main unit		
3	CD-ROM		The CD-ROM contains the following ● final inspection sheet ● instruction manual

2. Outline

The C12880MA is a spectrometer (polychromator) that combines a reflective grating and an image sensor in a compact package.

The C12880MA is equipped with a high-sensitivity CMOS linear image sensor while maintaining the ultra-compact size of the spectrometer realized by MEMS technology. The image sensor also has an electronic shutter function. This allows for a wider range of applications in various spectroscopic applications requiring weak light measurement, trigger measurement, etc.

The wavelength range is 340 to 850 nm (ultraviolet to visible wavelength), and the wavelength resolution is typ. 12 nm. In addition, having adopted a metal package with hermetic sealing, high reliability is provided against humidity effects, greatly improving the long term stability of our micro spectrometer product.



C12880MA

3. Setup

3-1. Marking



- ① Type index
B : C12880MA
- ② Individual identification number
- ③ QR Code (Data Matrix)
Coding the individual identification number



When reading QR Code (Data Matrix), use a barcode reader that meets the following specifications

Data Matrix Version	ECC200	--
Data Matrix Size*1	3.6 × 3.6	mm

*1:cell size 0.3mm□, 12x12 cell

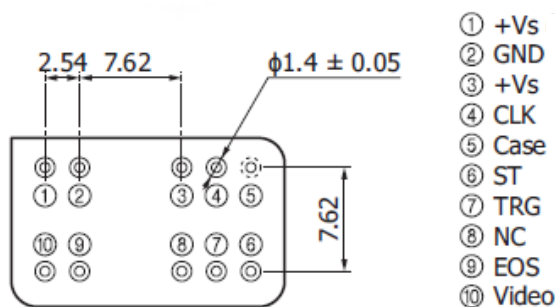
3-2. Pin connections

Leads are used to make electrical connections to external circuits.

Pin no.	Symbol	Name	I/O	Description
1	+Vs	Supply voltage	I	Power supply for sensor: 5V
2	GND	Ground	-	Sensor Ground
3	+Vs	Supply voltage	I	Power supply for sensor: 5V
4	CLK	Clock pulse	I	Sensor clock pulse
5	Case	Case	-	Case Connection
6	ST	Start pulse	I	Sensor start pulse
7	TRG	Trigger pulse	O	Pulse for capture sensor video signal
8	-	Fixing Pin	-	Do not connect electrically
9	EOS	End of scan	O	End of sensor scan
10	Video	Video output	O	Sensor video output

Note) Pin No. 5 and the case are at the same potential. Use the product in a condition where the case is not in contact with other potentials.

The part where the case contacts should be at the same potential as pin No. 5 or electrically insulated



C12880MA Bottom View

4. Connection

The C12880MA is a spectrometer that spectra incident light with optics, converts it to photoelectric signals with an image sensor, and outputs the signals as video signals. It is designed to be connected to a later-stage circuit that processes the output signal.



WARNING / 警告

- Since the products are precision optical products, excessive load applied to the case may cause the case to warp or break, the chip to peel off, wires to break, or unsatisfactory optical characteristics to be obtained. Please be very careful when mounting the product on a board or inserting it into a socket.

由于产品是精密光学产品，过大的负载可能导致外壳翘曲或断裂、芯片剥落、导线断裂或获得不满意的光学特性。请小心地将产品安装在木板上或插入插座。

- Do not handle the product in the following ways as it may cause destruction of the case.
请勿以以下方式处理该产品，因为它可能导致外壳损坏。

Repeatedly apply stress to the lead / 对铅反复施加压力

Perform rapid cooling and heating / 进行快速冷却和加热

4-1. Connection method

There are two methods of connection: (1) using sockets and (2) soldering. When making connections, please pay attention to the following information.

● Connection using sockets

- When mounting the product in the socket, check the pin connection positions. Reversing the mounting direction or misalignment of the product may result in damage to the product. Also, be sure to turn on the power supply after installing the product.
- Reverse insertion, incorrect insertion, or short-circuiting between terminals of the product may cause malfunction, so be very careful.
- Repeated application of external force to the leads may cause damage. When installing the product in a location where vibration is likely to occur, mount the product after securing it with resin or a holder, etc.

● Soldering connections

- Do not solder at high temperatures or for long periods of time, paying sufficient attention to the soldering iron tip temperature and soldering time. Soldering at high temperatures and for long periods of time may cause cracks in the package and peeling of the window material. Recommended soldering conditions are 350 °C or less for 3.5 seconds or less for hand soldering and 260 °C or less for 10 seconds or less for flow soldering. If these conditions are not met, pinch the base of the lead with tweezers or the like to dissipate heat so that heat is not transferred to the package portion of the product.
- During soldering, take measures to prevent dirt from adhering to the window material due to solder or flux splattering.
- Use no-clean solder or rosin-based flux. Use of flux with a certain degree of acidity or alkalinity or inorganic flux may cause lead corrosion.

4-2. About the drive circuit to be connected

When designing the circuit board for driving this product, pay attention to the following contents.

(4) Product mounting surface

- The product mounting surface of the circuit board is incorporated into an optical system for suppressing the input of external light. Therefore, adjustment is facilitated by mounting a variable resistor or the like on the back surface of the product mounting surface.

(5) Circuit board

- The mounting holes (on the circuit board) should be slightly larger than the screw diameter to allow for fine adjustment of the position of the light-receiving part. To prevent the focal point from being shifted, use a circuit board that is resistant to warping due to temperature changes.

(6) Circuit components

- When the temperature of the product changes, characteristics such as sensitivity and dark current will change. To suppress the temperature rise of the product, it is recommended to use circuit components that generate as little heat as possible. Also, keep heat-generating circuit components away from the product and take measures to dissipate heat.

(7) Ground

- To prevent noise contamination from the digital section to the analog section via ground, separate the ground of the digital and analog sections, and make the ground line thicker for low resistance.

(8) Digital signal

- Input clock pulse line has voltage fluctuation. Therefore, keep input clock pulse line as far as possible from video signal line and voltage supply line. In addition, apply input clock pulse of prescribed timing and voltage to product connection terminal on board. Pay special attention during high-speed operation.

(9) Analog signals

The wiring width and distance from the video output terminal to the amplifier should be as short as possible. For products with active video and dummy video, the signal lines on the active video side and the dummy video side should have the same wiring width and wiring distance to make the capacity as similar as possible. Furthermore, to avoid noise contamination of the output signal, the video signal line should be separated from the clock line and other digital signal lines with voltage variations, and the video and digital signal lines should not cross each other at the front and back of the circuit board. This reduces clock field-through.

(10) Supply voltage

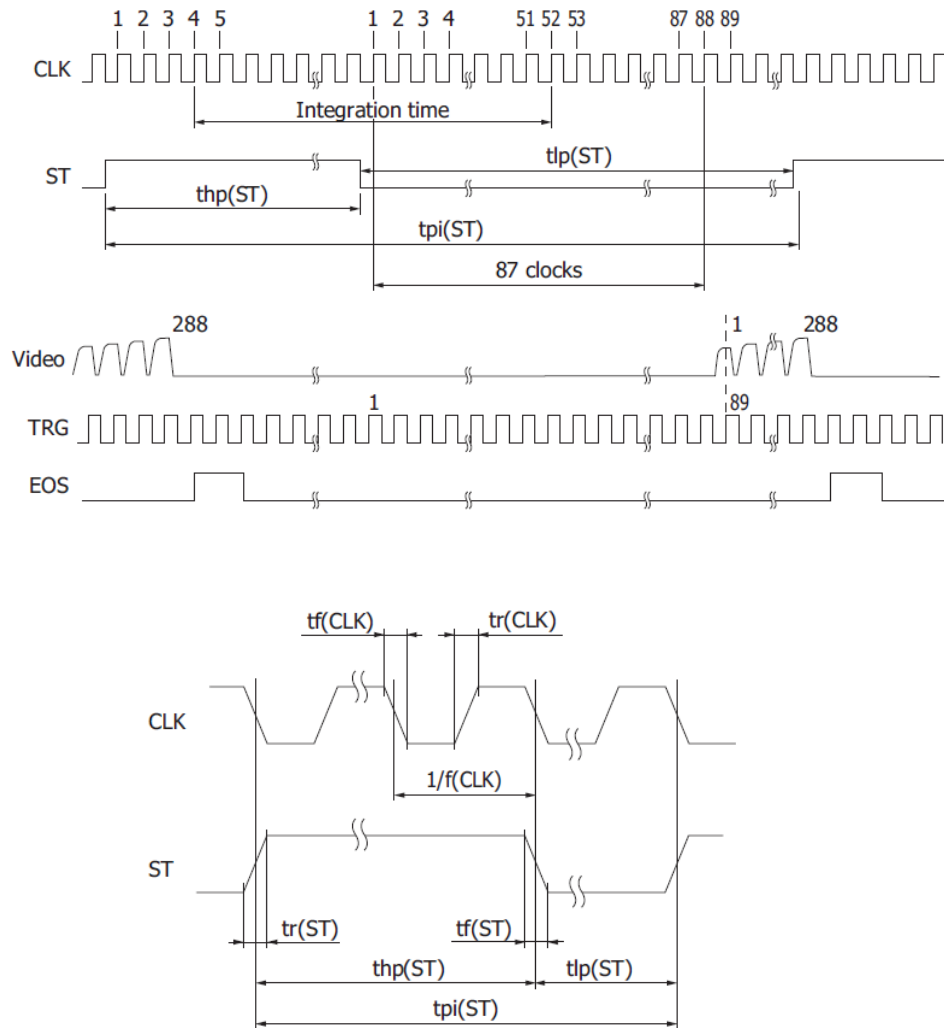
- If the supply voltage to the product fluctuates, the output characteristics will not be stable. Therefore, use a power supply with low voltage fluctuation and ensure that the supply voltage to the product is not affected by external voltage fluctuations. Also, ensure that the voltage supplied to the product does not vary due to voltage fluctuations in the power supply line caused by the operation of circuit components on the circuit board. Keep the power supply line to the product as far away as possible from digital signal lines with voltage variations, such as clock lines.

(11) Other

- Noise generated from mechanical drive units of the equipment in which the product and circuit board are incorporated may be mixed in with the output signal. Be sure to shield the circuit board.
- The characteristics may deteriorate when operating at high temperatures. Take measures to dissipate heat as necessary.
- Since the case is conductive, design patterns to avoid shorts due to contact with circuit patterns.

5. Operating Procedure

5-1. Timing chart



Parameter	Symbol	Min.	Typ.	Max.	Unit
Start pulse width interval ^{*1}	$tpi(ST)$	$381/f$	-	-	s
Start pulse high period ^{*2}	$thp(ST)$	$6/f(CLK)$	-	-	s
Start pulse low period	$tlp(ST)$	$375/f$	-	-	s
Start pulse rise and fall times	$tr(ST), tf(ST)$	0	10	30	ns
Clock pulse duty	-	45	50	55	%
Clock pulse rise and fall times	$tr(CLK), tf(CLK)$	0	10	30	ns

^{*1}: The shortest period required to output the video signals from all pixels.

^{*2}: The integration time equals the high period of ST plus 48 CLK cycles.

The shift register starts operation at the rising edge of CLK immediately after ST goes low.

The integration time can be changed by changing the ratio of the high and low periods of ST.

If the first TRG pulse after ST goes low is counted as the first pulse, the Video signal is acquired at the rising edge of the 89th TRG pulse.

Regarding the timing of ST start-up and start of accumulation

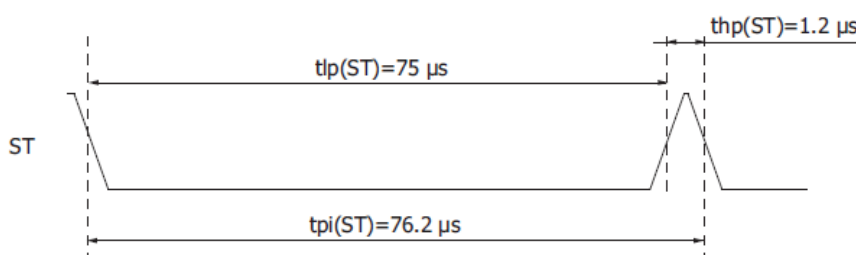
As shown in the timing chart above, the image sensor installed in this product has different timing for ST startup and timing for accumulation start. As an example of the timing chart, accumulation starts 4 CLKs after the rising edge of ST, but in reality, accumulation starts 1 CLK, 2 CLKs, 3 CLKs, or 4 CLKs after the rising edge of ST, depending on the number of CLKs input up to the rising edge of ST,

In reality, accumulation starts after 1, 2, 3, or 4 CLKs after the rising edge of ST, depending on the number of CLKs input up to the rising edge of ST. Therefore, it is recommended that the Start Pulse High and Start Pulse Low periods be set to $4 \times N$ (N is an integer) CLKs each to prevent variations in accumulation time during continuous repetitive measurement.

Operation example

This is an operating example when the clock pulse frequency is set to maximum (video data rate is also set to maximum), the time perscan to minimum, and the integration time to maximum.

- Clock pulse frequency $[f(\text{CLK})] = \text{Video data rate}$
 $= 5 \text{ MHz}$
- Start pulse cycle $[t_{pi}(\text{ST})] = 381/f(\text{CLK})$
 $= 381/5 \text{ MHz}$
 $= 76.2 \mu\text{s}$
- Low period of start pulse min. $[t_{lp}(\text{ST})] = 375/f(\text{CLK})$
 $= 375/5 \text{ MHz}$
 $= 75 \mu\text{s}$
- High period of start pulse $[t_{hp}(\text{ST})] = \text{Start pulse cycle } [t_{pi}(\text{ST})] - \text{Low period of start pulse min. } [t_{lp}(\text{ST})]$
 $= 76.2 \mu\text{s} - 75 \mu\text{s}$
 $= 1.2 \mu\text{s}$



KACCC0772EA

Integration time is equal to the high period of start pulse + 48 cycles of clock pulses, so it will be $1.2 \mu\text{s} + 9.6 \mu\text{s} = 10.8 \mu\text{s}$.

5-2. Wavelength conversion coefficient

Each product is assigned an individual Wavelength conversion coefficient.

By referring to the Wavelength conversion coefficient*1 on the final inspection sheet and applying it to the formula below, you can determine which wavelength each pixel is detecting.

$$\text{Wavelength[nm]} = A0 + B1 \times \text{pix} + B2 \times \text{pix}^2 + B3 \times \text{pix}^3 + B4 \times \text{pix}^4 + B5 \times \text{pix}^5$$

Note) pix: pixel number (1 to 288)

*1: Coefficient for converting the pixel No. of the image sensor to the wavelength; a coefficient for converting the count value after A/D conversion to a value proportional to the input light intensity is not provided.

6. Specification

6-1. Configuration

Item	Specification	Unit
Image Sensors	High-sensitivity CMOS linear image sensor with slit	-
Number of pixels	288	pixels
Pixel size (H × V)	14 × 200	μm
Slit* ¹ (H × V)	50 × 500	μm
NA* ²	0.22	-
Dimensions (W × D × H)	20.1 × 12.5 × 10.1	mm
Weight	5	g

*1: Size of incident aperture

*2: Number of apertures (solid angle)

6-2. Absolute maximum ratings

(If not specified Ta=25 °C)

Item	Symbol	Terms	Rated value	Unit
Supply voltage	Vs max		-0.3 to +6	V
Clock pulse voltage	V(CLK)		-0.3 to +6	V
Start pulse voltage	V(ST)		-0.3 to +6	V
Operating temperature	Topr	no condensation* ¹	+5 to +50	°C
Storage temperature	Tstg	no condensation* ¹	-20 to +70	°C

*1: In a humid environment, temperature differences between the product and its surroundings can easily cause condensation on the product surface, which may affect its characteristics and reliability. Note) Exceeding the absolute maximum ratings even momentarily may result in a loss of product quality. Always use the product within the absolute maximum ratings.

6-3. Recommended terminal voltage

(Ta=25 °C)

Item	Symbol	Min.	Typ.	Max	Unit
Supply voltage	Vs	4.75	5	5.25	V
Clock pulse voltage	Highlevel	3	Vs	Vs + 0.25	V
	Lowlevel	0	-	0.3	
Start pulse voltage	Highlevel	3	Vs	Vs + 0.25	V
	Lowlevel	0	-	0.3	

6-4. Electrical and optical characteristics

(Ta=25 °C, Vs=5 V, V(CLK)=V(ST)=5 V)

Item	Symbol	Terms	Min.	Typ.	Max	Unit
Clock pulse frequency	f(CLK)		0.2	-	5	MHz
Video Rate	VR		-	f(CLK)	-	Hz
Output impedance*1	Zo		-	150	-	Ω
Current Consumption	I	f(CLK)=5 MHz	-	20	-	mA
conversion efficiency	CE		-	50	-	μV/e ⁻
Dark output voltage	Vd	Accumulation Time =10 ms	-	0.8	8	mV
Saturation output voltage *2	Vsat		3.6	4.3	4.6	V
Readout noise	Nr		-	1.8	3.6	mV rms
Output offset voltage	Vo		0.3	0.5	0.9	V
Sensitivity wavelength range	λ		340	-	850	nm
Spectral resolution (FWHM)	-	λ=340 to 850nm	-	12	15	nm
Wavelength Reproducibility	λr	Constant incidence conditions, etc.	-0.5	-	+0.5	nm
Wavelength temperature dependence	λTd		-0.1	-	+0.1	nm/°C
Spectral stray light	SL	Enter emission line spectrum (655nm)	-	-	-25	dB

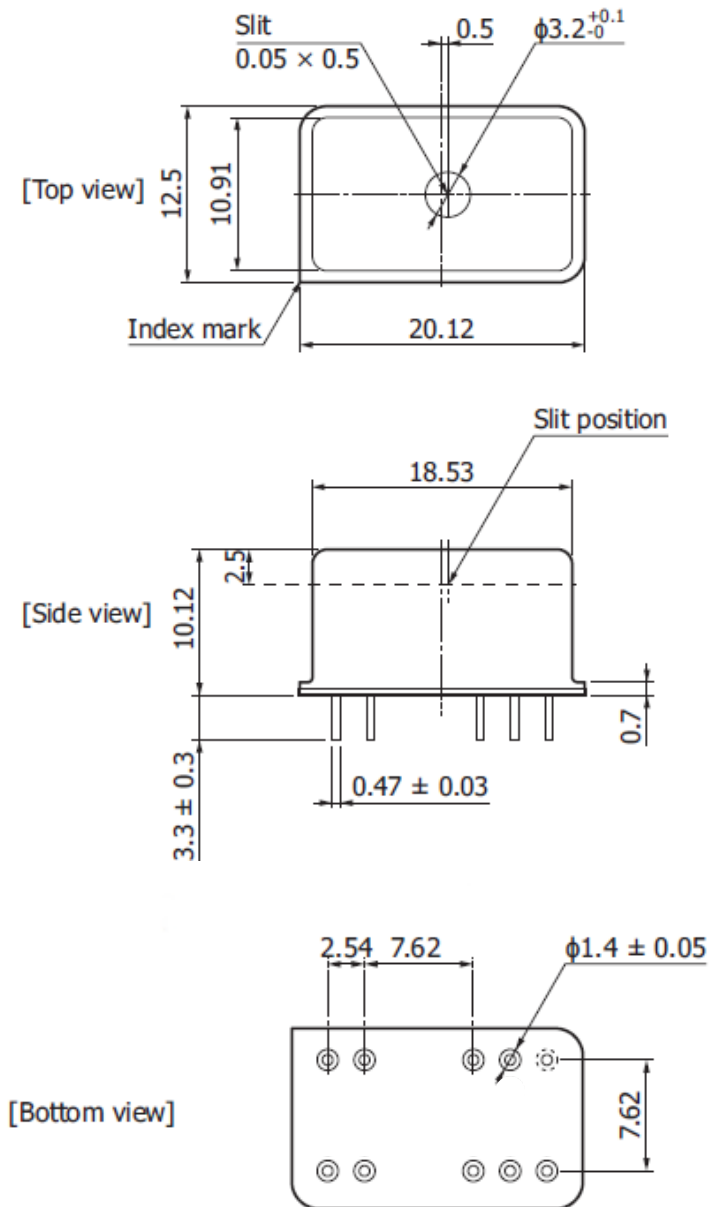
*1: Increased current consumption at the video output pin (pin No. 10) causes the image sensor chip temperature to rise and dark current to increase. A buffer amplifier should be connected to the video output pin to keep current flow as low as possible.

*2: Voltage difference between video signal output Video (pin No. 10) and output offset voltage Vo
Example: If the video signal output Video is 4.8 V and the output offset voltage Vo is 0.5 V, the saturation output voltage Vsat is 4.3 V.

*3: Ratio of the bright line spectrum (655 nm) versus the counts measured at that wavelength ±20 nm

7. External dimensions

Unit: mm, Tolerance without indication: ± 0.2



8. Warranty, Repair / 保修、维修

● Warranty / 保修

- When using the product described in this manual, observe the maximum rating and precautions. Our company makes every effort to improve the quality and reliability of this product but does not guarantee the integrity of this product. When the product is used for equipment that may infringe on human life, body or property, it is dangerous unless appropriate safety measures are taken in consideration of possible defects. Our company assumes no responsibility for such use unless the prior written consent of our company such as specifications has been obtained.

使用本说明书中描述的产品时，请遵守最大额定值和注意事项。公司尽一切努力提高本产品的质量和可靠性，但不保证本产品的完整性。当产品用于可能危害人员生命、身体或财产的设备时，除非考虑到可能存在的缺陷，采取适当的安全措施，否则十分危险。除非事先取得本公司的书面同意（例如规格说明），否则本公司对此类使用不承担任何责任。

- When providing guidance on the operation of the product to end users, the reader is requested to carefully explain the function, performance, and handling of the product and equipment using the product, and to give appropriate warnings and indications.

在向最终用户提供产品操作指导时，应仔细解释产品以及使用该产品的设备的功能、性能和处理，并给予适当的警告和提示。

- The warranty of the product in this document is limited to repair of the device or delivery of a substitute if the defect is found within 1 year after delivery and our company is notified thereof. Our specifications (Modification and violation of various conditions concerning the environment, application field, usage, storage, disposal, etc., described in this document), even within the warranty period.

本文档所述的产品保修仅限于在发货后 1 年内发现缺陷并通知我公司的情况下，对设备进行维修或交付替代品。即使在保修期内，本公司的规格（本文档中所述的环境、应用领域、使用、储存、处置等各种条件的修改和违反）也不受此限制。

- Please do not reproduce or produce the contents of this document without our company's permission.
- When disposing of this product, treat it as industrial waste according to the regulations of the national or local government.

在处置本产品时，请按照国家或地方政府的规定将产品作为工业废弃物进行处理。

● Repair / 维修

If you notice anything unusual, please contact Hamamatsu Photonics or your regional representative with the product name, serial number, and description of the fault (see the end of this document). We try to complete repairs as quickly as possible, but in the following circumstances, there is a possibility that it may be necessary to charge a repair fee or refuse repair:

如果您发现任何异常，请联系 Hamamatsu Photonics 或您所在地区的代表，并提供产品名称、序列号和故障描述（请参阅本文档末尾内容）。我们会尽力尽快完成维修，但在以下情况下，可能需要收取维修费用或拒绝维修：

- (1) **If a long period of time has passed since you purchased the product**
如果您购买本产品已经很长时间
- (2) **If repair parts are no longer produced**
如果维修所需零件已不再生产
- (3) **If you have mode modifications**
如果您有模式修改
- (4) **If the damage is considered to be substantial.**
如果认定存在实质性损害。
- (5) **When an abnormal phenomenon is not reproduced at Hamamatsu Photonics**
当 Hamamatsu Photonics 无法再现异常现象时
- (6) **When it is caused by other devices used at the same time**
因同时使用其他设备而引起的故障

Revision history

Revision date	Revision	Revision details
2015/2/24	-	First issue
2015/4/27	A	The type number turns into C12880MA from C12880MA(X)
2017/2/28	B	Renaming of pin connections
2018/7/9	C	Revision of timing chart
2020/2/28	D	Revision of the value of recommended terminal voltage
2020/5/19	E	Revision of recommended terminal voltage of V(ST) and V(CLK)
2025/06/11	F	Change of format and correction of errors
2025/12/5	G	Addition of Chinese text to caution/warning notice

For questions or consultations about our products, please use the inquiry form or call the nearest sales office.

Inquiry: <https://www.hamamatsu.com/all/en/support/inquiry.html>

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